

FORT RELIANCE, NT, Canada

WMO: 711600

Lat: 62.711N

Lon: 109.168W

Elev: 168

StdP: 99.33

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.1	-38.8	-45.8	0.0	-41.0	-43.5	0.1	-38.7	9.1	-19.6	8.1	-20.4	1.5	210	0.759

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.6	24.6	15.8	22.5	14.7	20.6	14.0	16.7	22.4	15.7	20.7	14.6	19.2	2.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
14.5	10.5	18.2	13.4	9.8	17.6	12.3	9.1	16.9	47.6	22.7	44.5	20.8	41.6	19.2	21.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	7.6	6.4	-43.2	27.9	2.0	2.1	-44.7	29.4	-45.9	30.6	-47.0	31.8	-48.5	33.3		
			-43.2	17.8	1.8	1.1	-44.4	18.6	-45.5	19.2	-46.5	19.8	-47.8	20.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.1	-25.9	-24.3	-18.8	-8.7	2.4	10.8	15.1	13.8	7.7	-0.9	-12.3	-21.4
	DBStd	16.05	8.03	8.03	9.38	8.21	6.03	4.42	3.60	3.53	4.26	4.89	6.55	8.06
	HDD10.0	5901	1114	962	894	562	241	44	3	7	92	338	670	973
	HDD18.3	8577	1373	1195	1152	811	493	227	112	147	318	596	920	1231
	CDD10.0	384	0	0	0	0	6	68	161	123	24	1	0	0
	CDD18.3	19	0	0	0	0	0	1	12	6	0	0	0	0
	CDH23.3	121	0	0	0	0	0	17	78	27	0	0	0	0
	CDH26.7	17	0	0	0	0	0	1	12	4	0	0	0	0
	WSAvg	2.8	2.2	2.3	2.5	2.8	2.8	2.5	2.3	2.6	3.3	3.8	3.6	2.4
Precipitation	PrecAvg	283	12	11	10	14	18	32	38	46	35	30	20	14
	PrecMax	388	26	29	34	30	47	118	100	90	76	74	49	36
	PrecMin	179	2	1	4	4	0	6	2	12	5	6	6	3
	PrecStd	56	5	7	7	8	11	25	23	23	15	15	10	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-3.9	-3.4	4.9	11.9	19.9	25.5	28.5	26.1	20.1	11.4	2.7	-2.4
		MCWB	-4.5	-4.3	1.7	6.2	10.6	15.1	17.2	17.1	13.6	8.0	1.3	-3.1
	2%	DB	-7.5	-6.9	1.6	8.0	16.4	22.8	25.6	23.2	17.6	8.5	0.7	-4.9
		MCWB	-7.9	-7.5	-1.0	3.5	8.7	13.7	16.3	15.7	12.4	6.1	-0.3	-5.6
	5%	DB	-10.8	-9.7	-1.5	5.2	13.3	20.5	23.3	21.2	15.4	6.7	-1.0	-7.3
		MCWB	-11.3	-10.3	-3.3	1.8	7.2	12.7	15.3	15.0	11.3	5.1	-1.9	-7.9
	10%	DB	-13.6	-12.8	-4.4	2.9	10.6	18.3	21.1	19.3	13.5	5.2	-2.8	-9.7
		MCWB	-14.1	-13.3	-5.8	0.5	5.7	11.6	14.3	14.1	10.3	3.8	-3.6	-10.2
	0.4%	WB	-4.6	-4.4	2.2	6.3	11.0	16.2	18.5	18.0	14.3	8.3	1.6	-3.2
		MCDB	-4.0	-3.6	4.7	11.7	19.4	22.9	25.5	23.4	18.7	10.8	2.5	-2.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-8.1	-7.5	-0.8	3.6	9.0	14.5	17.0	16.6	12.8	6.3	-0.2	-5.5
		MCDB	-7.7	-6.8	1.5	7.5	15.8	20.8	23.3	21.6	16.6	8.0	0.7	-4.9
	5%	WB	-11.5	-10.3	-3.2	2.0	7.4	13.3	16.0	15.6	11.6	5.2	-1.9	-7.8
		MCDB	-11.1	-9.8	-1.6	4.9	12.7	19.4	21.5	20.1	14.8	6.7	-1.1	-7.3
	10%	WB	-14.4	-13.3	-5.9	0.6	5.8	12.1	14.9	14.6	10.5	3.8	-3.6	-10.2
		MCDB	-14.0	-12.8	-4.5	2.7	10.2	17.7	19.8	18.5	13.2	5.1	-2.8	-9.7
Mean Daily Temperature Range	5% DB	MDBR	7.9	9.1	11.8	12.2	10.6	10.7	9.6	7.9	6.2	4.5	5.9	7.1
		MCDBR	8.7	10.4	13.3	12.7	14.1	14.0	12.4	11.0	9.1	6.1	5.4	7.9
	5% WB	MCWBR	8.3	9.9	11.3	8.7	8.2	7.2	5.5	5.1	5.0	4.3	5.2	7.7
		MCDWR	8.5	10.4	13.3	12.1	13.4	13.0	10.9	9.6	8.4	5.9	5.7	8.0
		MCWBR	8.4	9.9	11.2	8.4	7.9	7.2	5.6	5.0	5.0	4.3	5.5	7.8
Clear-Sky Solar Irradiance	taub		0.181	0.238	0.254	0.309	0.328	0.349	0.368	0.352	0.310	0.268	0.191	0.147
	taud		2.026	2.211	2.275	2.185	2.383	2.423	2.413	2.447	2.554	2.404	2.018	1.942
	Ebn at Noon		518	738	869	873	889	873	844	826	807	699	472	356
	Edh at Noon		31	60	83	114	103	102	100	89	66	50	30	16
All-Sky Solar Radiation	RadAvg		0.27	1.09	2.85	5.02	5.89	5.93	5.29	3.91	2.32	0.89	0.28	0.12
	RadStd		0.02	0.08	0.15	0.26	0.41	0.41	0.30	0.22	0.20	0.12	0.04	0.01

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A

Nomenclature: See separate page